OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

12.9 Junction reference line

Junction reference lines are geometric references related to junctions. A junction reference line is defined in the reference line coordinate system. A junction reference line attaches the `<elevationGrid>` element and other relevant objects to the junction, for example traffic islands or potholes. A junction reference line has no lanes and no elevation or superelevation.

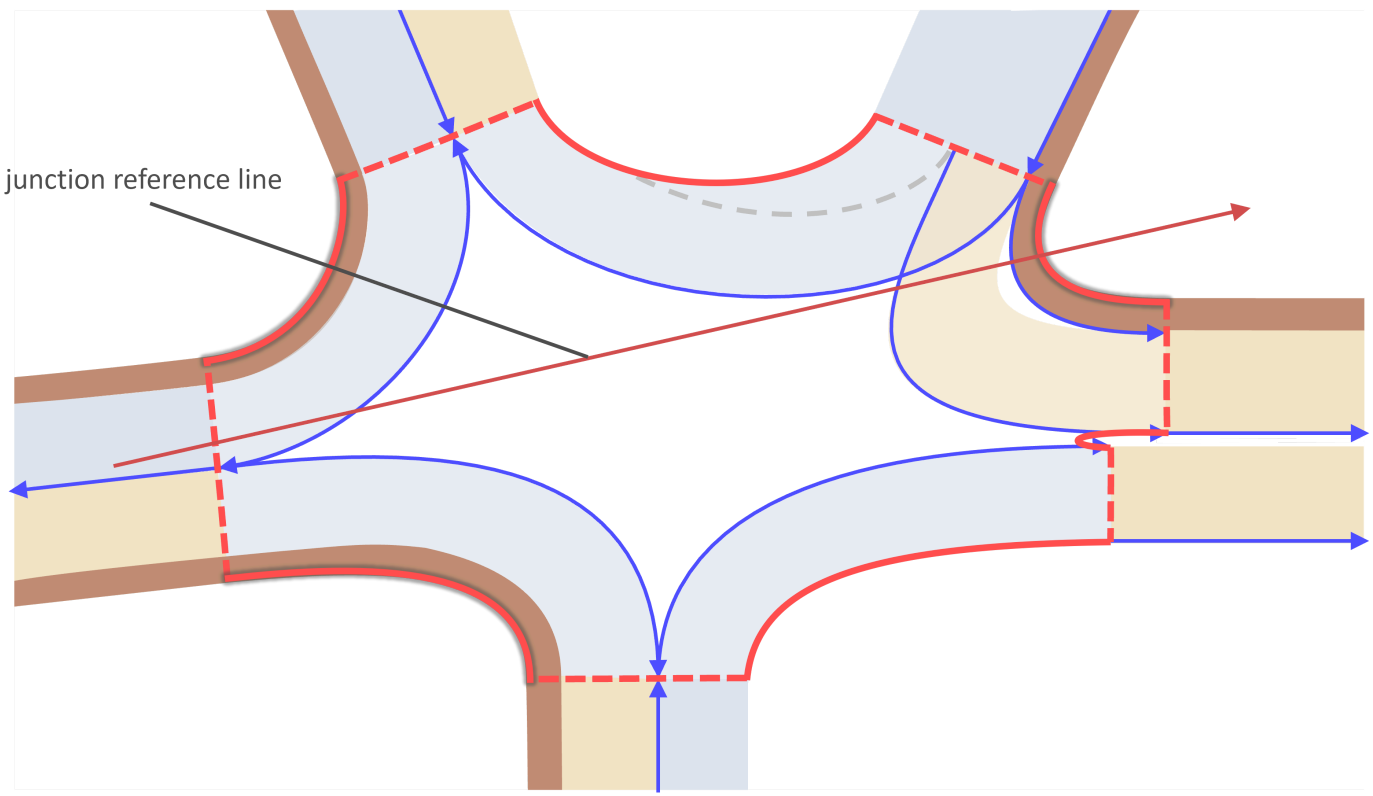

Figure 95. Junction reference line ~red arrow~

Figure 95 shows a junction and a junction reference line.

In ASAM OpenDRIVE, junction reference lines are represented by the `<planView>` element within the `<junction>` element.

```
UML class: t_road_planView
XML tag:   <planView> (Multiplicity: 1)
```

Contains geometry elements that define the layout of the road reference line in the x/y-plane (plan view).

`<geometry>` element

In ASAM OpenDRIVE, the geometry of a junction reference line is represented by the `<geometry>` element within the `<planView>` element.

```
UML class: t_road_planView_geometry
XML tag:   <geometry> (Multiplicity: 1..*)
```

Table 76. Attributes of the <geometry> element

Name	Type	Use	Unit	Description
`hdg`	double	required	rad	Start orientation (inertial heading)
`length`	t_grZero	required	m	Length of the element's reference line
`s`	t_grEqZero	required	m	s-coordinate of start position
`x`	double	required	m	Start position (x inertial)
`y`	double	required	m	Start position (y inertial)

The `<planView>` element contains one `<geometry>` element with the `<line>` element.

XML example

```
<junction id="15">
    <planView>
        <geometry s="0.0000000000000000e+00"
                  x="3.0505642010572856e+01"
                  y="3.6353713565101820e+01"
                  hdg="2.9842438203363086e-01"
                  length="7.5257400000000004e+01">
            <line/>
        </geometry>
    </planView>
    <object type="roadSurface" subtype="pothole" name="" id="1" s="3.1064163564293011e+01" t="1.6886219784199805e+0
        <surface>
          <CRG file="pothole.crg" hideRoadSurfaceCRG="true" zScale="1"/>
        </surface>
      </object>
</junction>
```

Rules

The following rules apply to junction reference lines:

- asam.net:xodr:1.8.0:junctions.geometry.only_one_line_element: Junction reference lines shall be defined by one `<geometry>` element. This `<geometry>` element shall have only one `<line>` element.
- asam.net:xodr:1.8.0:junctions.geometry.ref_line_definition: The `<geometry>` element of a junction reference line shall be defined in a way that every point of the junction can be reached with a perpendicular straight line.

- asam.net:xodr:1.8.0:junctions.geometry.correct_junction_boundry: If a junction boundary is specified, a junction reference line shall cross the junction boundary or be at least tangent to the junction boundary at one point.

Related topics

- Section 12.11, "Junction elevation grid"
- Section 12.10, "Junction boundary"